

FlexForge — How to Run

Step-by-step guide for running the engine locally, on a real database, and as a container · v0.1.0

0. Prerequisites

- **JDK 21** (check: `java -version`)
- **Maven 3.9+** (check: `mvn -version`)
- Optional: Docker (for container builds), Postgres/MySQL (for a real DB)
- No database needed to start — the engine defaults to in-memory H2.

1. Get the code

The project lives under `flexforge/` on the branch `claude/backend-architecture-plan-xcgh3m`.

```
git clone https://github.com/0101vivek/0101vivek.git
cd 0101vivek
git checkout claude/backend-architecture-plan-xcgh3m
cd flexforge
```

2. Run it (zero setup — in-memory H2)

```
mvn spring-boot:run
```

You should see logs ending with:

```
[engine] config 'crm-demo' v0.1.0 validated: 4 entities, REST=true, GraphQL=true
[schema] creating table customer ...
[engine] schema ready - app 'crm-demo' is live
Tomcat started on port 8080
```

Then open the admin UI in a browser:

```
http://localhost:8080/
```

3. Try the APIs (in another terminal)

REST:

```
curl -s -X POST localhost:8080/api/Customer -H 'Content-Type: application/json' \
  -d '{"email":"ada@example.com","fullName":"Ada Lovelace","tier":"gold"}'
```

```
curl -s localhost:8080/api/Customer
curl -s 'localhost:8080/api/Customer?tier_like=gol'
```

GraphQL:

```
curl -s -X POST localhost:8080/graphql -H 'Content-Type: application/json' \
  -d '{"query":{"{ customerList { id email tier } }"}}'
```

A no-code flow:

```
curl -s -X POST localhost:8080/flows/grade/run -H 'Content-Type: application/json' \
  -d '{"score":70}' # -> {"result":"pass"}
```

Real-time + audit:

```
curl -s localhost:8080/realtime/recent # change events
curl -s localhost:8080/__audit # audit log
curl -s localhost:8080/__meta/openapi.json # OpenAPI spec
```

4. Point at a real database (same image, no code change)

Connection details come from environment variables (12-factor):

```
export DB_ENGINE=postgres
export DB_URL='jdbc:postgresql://localhost:5432/crm'
export DB_USERNAME=crm
export DB_PASSWORD=secret
mvn spring-boot:run

# MySQL: DB_ENGINE=mysql, DB_URL=jdbc:mysql://localhost:3306/crm
```

5. Run a different app with the same engine

```
export FLEXFORGE_CONFIG_PATH=/path/to/your/app.yaml
mvn spring-boot:run
```

6. Build artifacts

```
mvn test          # run the 30 integration tests on H2
mvn package       # build the fat jar (target/flexforge-engine-0.1.0.jar)
java -jar target/flexforge-engine-0.1.0.jar

# OCI image with no Dockerfile and no source in the image:
mvn compile jib:build -Dimage=ghcr.io/you/flexforge-engine:0.1.0
```

7. Enable security (optional)

Set `security.enabled: true` in the config with users + role rules, then:

```
TOKEN=$(curl -s -X POST localhost:8080/auth/login -H 'Content-Type: application/json' \
  -d '{"username":"admin","password":"admin123"}' | jq -r .token)
curl -s localhost:8080/api/Customer -H "Authorization: Bearer $TOKEN"
```

8. Troubleshooting

Symptom	Fix
Port 8080 in use	Set <code>server.port</code> or stop the other process
'database.url is required'	Set <code>DB_URL/DB_ENGINE</code> , or unset them to use H2
Config invalid on boot	Read the printed JSON-Schema/semantic errors; fix <code>app.yaml</code>
401 on API calls	Security is on — log in and send the Bearer token
Java version error	Install/select JDK 21